EVERY MODULE IS A QUOTIENT OF A FLAT MODULE, INDEED OF A DIRECT SUM OF SHEAVES PUSHED FORWARD FROM OPENS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-modules/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-modules/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

Editorial note. Bibliography provenance: neither the counted statement nor its proof contains a citation (this lane rescans the whole span before writing the card), so the card carries no bibliography and no entry was reconstructed.

Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The adjunction describing morphisms from a module extended by zero from an open into a module as sections of the restriction. Used at the start of the counted proof (source lines 2868-2871). Produces one map into the given module for every section over every open.
- The flatness of the module extended by zero from an open, and the flatness of colimits of flat modules. Invoked at the end of the counted proof (source lines 2878-2879) as pointers to two lemmas of this chapter. Shows that the direct sum of those extended modules is flat.

Source reference. Original lemma lemma-colimits-flat: sources/stacks-modules/pinned-original-source.tex, source lines 2879–2879. Navigation pointer only: the counted text refers to the Stacks item lemma-colimits-flat, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-j-shriek-flat: sources/stacks-modules/pinned-original-source.tex, source lines 2879–2879. Navigation pointer only: the counted text refers to the Stacks item lemma-j-shriek-flat, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space.*

(1) *Any sheaf of $\mathcal{O}_X$-modules is a quotient of a direct sum $\bigoplus j_{U_i!}\mathcal{O}_{U_i}$.*

Date: stacks-project:modules.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(2) *Any $\mathcal{O}_X$-module is a quotient of a flat $\mathcal{O}_X$-module.*

Proof. Let $\mathcal{F}$ be an $\mathcal{O}_X$-module. For every open $U \subset X$ and every $s \in \mathcal{F}(U)$ we get a morphism $j_{U!}\mathcal{O}_U \to \mathcal{F}$, namely the adjoint to the morphism $\mathcal{O}_U \to \mathcal{F}|_U$, $1 \mapsto s$. Clearly the map

$$\bigoplus\nolimits_{(U,s)} j_{U!}\mathcal{O}_U \longrightarrow \mathcal{F}$$

is surjective, and the source is flat by combining Lemmas lemma-colimits-flat and lemma-j-shriek-flat. □